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**EFFECT OF FEATURE SELECTION ON CLASSIFICATION
ACCURACY ON HIGH-DIMENSIONAL GENE EXPRESSION DATA**

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Abstract

Gene expression data derived from RNA sequencing technologies typically consist of thousands of gene features measured on a relatively small number of biological samples. Such high-dimensional datasets present significant challenges for machine learning models, including overfitting, increased computational complexity, and reduced generalization performance. Feature selection techniques are commonly used to mitigate these challenges by identifying the most informative genes for classification tasks. However, the comparative impact of different feature selection methods on classification accuracy remains an open question in many high-dimensional biological datasets. This study aims to investigate how various feature selection approaches influence classification performance in bulk RNA-seq gene expression data. By applying filter, wrapper, and embedded feature selection methods and evaluating their effects using a consistent machine learning model, this research seeks to provide insight into dimensionality reduction strategies for improved predictive modeling in genomics.

Introduction

Inside every cell, DNA contains genes that serve as instructions for cellular activity. However, not all genes are active at the same time. When a gene is active, the cell produces RNA from it, and that level of activity is termed *gene expression*. In essence, gene expression answers the question: “How active is this gene in this sample?”

Gene expression refers to the process by which genetic information encoded in DNA is transcribed into RNA, ultimately influencing protein production and cellular function [?]. In disease studies, gene expression profiling is widely used to identify molecular differences between healthy and diseased samples. Advances in high-throughput sequencing technologies, particularly RNA sequencing, have enabled large-scale measurement of gene expression levels across biological samples.

A typical gene expression dataset consists of thousands of genes (features) and a very small number of samples. This imbalance creates what is commonly known as the “high-dimensional, low-sample-size” problem. In such settings, machine learning algorithms often struggle to generalize effectively due to overfitting and the curse of dimensionality. Consequently, dimensionality reduction techniques, particularly feature selection methods, are essential for improving model performance and interoperability.

Feature selection aims to identify a subset of relevant genes that contribute most significantly to classification tasks, such as distinguishing between disease and healthy samples. Despite the widespread use of feature selection techniques, there remains a need for systematic evaluation of their relative effectiveness in improving classification accuracy on high-dimensional gene expression data.

This project investigates the role of gene expression data in disease prediction, with a focus to explore various feature selection techniques and evaluate their impact on machine learning classification accuracy.

Problem Statement

Gene expression datasets contain thousands of gene features but relatively few samples. This high dimensionality can negatively affect classification performance by increasing model variance, promoting overfitting, introducing noise from irrelevant or redundant features, and poor generalization in classification tasks. While feature selection is commonly used to address this issue, it is not always clear which feature selection techniques most effectively improve classification accuracy on high-dimensional gene expression data.

High-Dimensional Data

High-dimensional data refers to datasets where each observation (sample) is described by a large number of variables (features, dimensions, or explanatory variables) relative to what is typical or convenient for analysis. In the simplest terms, it means data with a large number of attributes or features — for example, thousands of genes measured across only a few samples.

The “Small n , Large p ” Problem

High-dimensional data occurs when the number of features (p) greatly exceeds the number of samples (n). In genomic datasets, this typically means:

- **High p :** Thousands of genes (dimensions).
- **Small n :** Often fewer than 100 clinical samples.

The Curse of Dimensionality: With too many features, distances between data points become meaningless, and models can find “patterns” in pure noise, leading to overfitting.

Why High-Dimensional Data is a Problem

Machine learning models trained on high-dimensional data commonly suffer from the following issues:

- **Overfitting:** The model memorizes noise instead of learning genuine patterns.
- **Curse of Dimensionality:** Too many features cause distances between points to become meaningless.
- **Slow Training:** Unnecessary information makes the training process slow and often inaccurate.

Feature Selection

Feature selection involves identifying the most important genes for prediction. It serves to reduce noise, improve model performance, prevent overfitting, and enhance the interpretability of results.

Types of Feature Selection Methods

0.0.1 Filter Methods

Filter methods use statistical techniques to evaluate which genes differ most between disease and healthy samples, independently of any learning algorithm. Examples include the t-test, ANOVA, correlation analysis, and mutual information.

Univariate Analysis. Each gene is evaluated independently using measures such as:

- **Variance Thresholding:** Genes with near-zero variance are removed.
- **Differential Expression (DE):** Genes are ranked by a two-sample t-statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Limitation: Filter methods ignore gene-gene interactions (epistasis).

0.0.2 Wrapper Methods

Wrapper methods use a machine learning model to evaluate subsets of genes. A typical example is Recursive Feature Elimination (RFE), which follows this iterative process:

Try gene subset $\rightarrow$ Train model $\rightarrow$ Check accuracy $\rightarrow$ Repeat

Recursive Feature Elimination (RFE). The RFE mechanism proceeds as follows:

1. Train a model (SVM or Random Forest) on all genes.
2. Rank genes by importance (weights or Gini impurity).
3. Remove the k least important genes.
4. Repeat until the optimal subset size is reached.

RFE captures inter-gene dependencies but is computationally expensive.

0.0.3 Embedded Methods

In embedded methods, feature selection occurs as part of the model training process. The model itself decides which genes are important. Common examples include LASSO and Random Forest feature importance.

L1 Regularization (LASSO). LASSO (Least Absolute Shrinkage and Selection Operator) is the industry standard for embedded feature selection:

$$\min_{\beta} \{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

The L_1 penalty forces coefficients of irrelevant genes to exactly **zero**, effectively selecting a sparse gene list. The hyperparameter λ controls the degree of sparsity.

Classification (Prediction Stage)

After feature selection, at least one classification model is trained using the selected features. Candidate models include Support Vector Machines (SVM), Logistic Regression, Random Forest, Convolutional Neural Networks (CNN), Artificial Neural Networks (ANN), and Naive Bayes. The goal is to predict disease versus healthy status, evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

Aim and Objectives

0.1 Aim

The aim of this study is to investigate the effect of different feature selection techniques on classification accuracy in high-dimensional gene expression datasets.

0.2 Objectives

- To analyze the structure and characteristics of bulk RNA-seq gene expression data.
- To implement various feature selection methods, including filter, wrapper, and embedded approaches.
- To train a machine learning classification model using features selected by each method.
- To evaluate and compare classification performance using appropriate metrics.
- To assess the extent to which dimensionality reduction improves model generalization.

Workflow

The project follows this sequence of steps:

1. Obtain a gene expression dataset.
2. Preprocess and normalize the data.
3. Apply feature selection methods.
4. Train a classification model.
5. Compare classification performance across methods.

Expected Outcomes

By the end of this project, we expect to:

- Identify the most effective feature selection techniques for gene expression data.
- Improve classification accuracy relative to using all features.
- Achieve meaningful reduction in the dimensionality of gene expression data.

Conclusion

Gene expression data is inherently high-dimensional, presenting significant challenges for machine learning models. Feature selection is essential for effective and interpretable modeling in this domain. This study focuses on quantifying the impact of different feature selection strategies on classification performance, with the aim of identifying best practices for genomic data analysis.